Bayes-Risk Theory of Task-Posterior Diffusion for Sparse Multimodal Remote Sensing Detection

Wentao Zhang, *Member, IEEE*, and Wentao Mo, *Member, IEEE*

***Abstract*—This article studies a formal Bayes-risk version of the open problem identified by Liu et al. (2024) (TGRS): task specific, end-to-end, and multimodal remote sensing diffusion for interpretation tasks. Existing diffusion pipelines often learn an image distribution or image score and attach a detector afterward, but this separation is not well matched to small sparse object detection, where target evidence can be weak in the main image, distributed across auxiliary modalities, or hidden by foreground–background imbalance. We study a decision-calibrated task-posterior diffusion formulation for this setting. Detection is modeled on a finite set of valid anchors, and a class-balanced risk is used to avoid the trivial all-background solution. Under a local Gaussian model with whitened image anchors and auxiliary modal features, the image-only and multimodal risks have closed forms. The image-only risk depends on diffused image contrast, while the multimodal task-posterior risk also contains an auxiliary Mahalanobis information term. Any nonzero whitened auxiliary contrast gives a strict Bayes-risk improvement. When the main-image contrast vanishes, the image-only detector tends to random guessing, while the multimodal detector remains nondegenerate if auxiliary information persists. We also prove population consistency, characterize the decision-calibrated posterior score, and give stability margins for posterior error and total-variation deviations. Without Gaussianity or conditional independence, a distribution-free total-variation result still guarantees that adding modalities cannot increase the optimal balanced risk. On real VEDAI RGB–NIR anchors, a decision-score Gaussian Q–Q R^2 of 0.929 and full-covariance closed-form/held-out errors of 0.062/0.070 support the approximation's decision-level utility; RGB alone gives 0.092/0.093.**

***Index Terms*—Bayes risk, class imbalance, diffusion models, multimodal remote sensing, score matching, small object detection, task posterior.**

I. Introduction

REMOTE sensing object detection often has a mismatch between the measured signal and the decision that must be made. Targets can occupy only a few pixels, appear sparsely in large scenes, have arbitrary orientations, and show weak contrast in the main optical or SAR image. Remote sensing systems also provide heterogeneous side information, including optical bands, SAR, infrared channels, elevation priors, metadata, and temporal observations. A method that improves or generates an image before asking whether a target exists may optimize a quantity that is only indirectly related to the detection decision.

The open problem addressed here is a formal Bayes-risk version of the direction identified by Liu et al. [1] in their 2024 survey of remote sensing diffusion models (TGRS): existing interpretation pipelines often use diffusion as a feature extractor and then attach a classifier or detector, a separation that can be suboptimal because the extracted features may not be fully suited to the task; they therefore point to task specific, end-to-end diffusion models and effective multimodal remote sensing diffusion as future research directions [1].

Diffusion models have had a strong impact on generative modeling. Nonequilibrium diffusion models, denoising diffusion probabilistic models, and score-based formulations show that estimating the score of a perturbed data distribution can support high-quality sampling and inverse problems [2], [3], [4]. These ideas have quickly entered remote sensing, with work on controllable image generation, text-to-image benchmarks, diffusion-based data generation for object detection, and diffusion features for semantic interpretation [5], [6], [7], [8]. This motivates the theoretical question studied below: if the image score is learned correctly, is small-object detection also optimal?

The answer is no in general. The image score $\nabla_x \log p_t(x \mid z)$ describes a noisy image distribution conditioned on auxiliary information, where $p_t(x \mid z)$ is the diffusion-time conditional image density and ∇_x is the gradient with respect to x. It does not by itself give the class-balanced posterior $\mathbb{P}(M_i = 1 \mid X_t, Z)$ needed for a sparse target decision at anchor i. In a simple limiting case, the image distribution can be independent of the target label while an auxiliary modality remains informative. A detector based only on the image or on the pure image score is then no better than random guessing, whereas a multimodal posterior detector can still do better.

We formalize this mismatch through a Bayes-risk comparison. The classical image-postprocessing paradigm is represented by detectors that observe only the local noisy image feature $V_{i,t}^{(0)}$. The task-posterior paradigm observes the local multimodal statistic $V_{i,t} = (V_{i,t}^{(0)}, V_i^{(1)}, \ldots, V_i^{(K)})$ and learns a class-balanced posterior head jointly with the diffusion score head. Under a local Gaussian model, the optimal image-only risk is

$$R^*_{i,\mathrm{img}}(t) = \Phi\left(-\frac{1}{2}\sqrt{\frac{a_t^2\|u_i\|_2^2}{a_t^2\sigma_{b,i}^2 + \sigma_t^2}}\right) \tag{1}$$

where Φ is the standard normal cumulative distribution function, u_i is the clean main-image target contrast, a_t is the signal

Received 6 June 2026; revised 2 September 2026; accepted 28 September 2026. Date of publication 8 October 2026; date of current version 19 October 2026. *(Corresponding author: Wentao Zhang.)*

The authors are with Tsinghua Shenzhen International Graduate School, Tsinghua University, Shenzhen 518055, China (e-mail: zhang-wt24@mails.tsinghua.edu.cn; mow10@mails.tsinghua.edu.cn).

This article has supplementary downloadable material available at https://doi.org/10.1109/TGRS.2026.3741532, provided by the authors.

Digital Object Identifier 10.1109/TGRS.2026.3741532

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retention, σ_t is the diffusion noise level, and $\sigma_{b,i}^2$ is the local background variance. The optimal multimodal risk is

$$R_{i,\text{all}}^*(t) = \Phi\left(-\frac{1}{2}\sqrt{\frac{a_t^2\|u_i\|_2^2}{a_t^2\sigma_{b,i}^2+\sigma_t^2}+\sum_{k=1}^{K}\delta_{ik}^\top\Sigma_{ik}^{-1}\delta_{ik}}\right) \quad (2)$$

where K is the number of auxiliary modalities, δ_{ik} is the class-conditional mean shift of modality k, and Σ_{ik} is its within-class covariance; $\delta_{ik}^\top\Sigma_{ik}^{-1}\delta_{ik}$ is the corresponding whitened detection information. The formula shows the source of the gain directly: any nonzero auxiliary Mahalanobis contrast strictly lowers the Bayes risk.

This work addresses three coupled challenges. First, sparse targets make raw empirical error compatible with a nearly all-background solution, so the decision criterion must expose foreground misses. Second, an image or representation score models an observation distribution but need not identify the target posterior used by the final detector. Third, conventional fusion can concatenate or align modalities without determining whether the added information lowers task risk. Existing sparse multimodal detectors and foundation-model pipelines primarily improve feature extraction, fusion, pretraining transfer, or domain adaptation; these are important representation mechanisms, but by themselves they provide neither a class-balanced decision target nor a theorem separating image-only and multimodal Bayes risks.

The contribution of this article is a risk-level account of this separation. We introduce a decision-calibrated task-posterior diffusion formulation for finite-anchor remote sensing detection under class-balanced risk, derive closed-form risks for image-postprocessing diffusion and multimodal task-posterior diffusion, and prove a strict gap whenever the auxiliary modalities contain nonzero whitened target information. We also prove that, as the main-image target contrast tends to zero, the image-only risk tends to $1/2$, while the multimodal task-posterior risk remains below $1/2$ if auxiliary information persists. The population objective combines denoising score matching with valid-anchor class-balanced cross-entropy, and its consistency holds at the diffusion times certified by the training time measure. The analysis is completed by posterior approximation margins, total-variation stability, a negative result for pure image-score diffusion, a top-s posterior decoding rule for sparse masks, and numerical checks under the analytic model.

The fundamental difference is therefore the optimized statistical object rather than a particular backbone: Remote sensing task-posterior diffusion (RSTPD) explicitly learns the diffusion time, class-balanced task posterior, and evaluates it by decision risk. A detector, multimodal encoder, or remote sensing foundation model may instantiate the localization map Γ_i, but the proposed theory specifies what posterior its head must recover, when an auxiliary modality yields strict improvement, and what remains true under arbitrary class-conditional distributions. The framework is consequently complementary to, rather than a replacement for, modern multimodal encoders. The scope of the Gaussian calculation is assessed in two complementary ways. First, a distribution-free result expresses balanced Bayes error through total variation and establishes weak multimodal dominance for arbitrary class-conditional laws and cross-modal dependence. Second, a tenfold scene-disjoint diagnostic on the VEDAI RGB–NIR small-vehicle benchmark evaluates decision-score Gaussian alignment, closed-form risk accuracy, and the empirical benefits of both joint-covariance modeling and flexible non-Gaussian posteriors.

This is a theory paper. It identifies a regime in which multimodal task-posterior diffusion is provably preferable to image-generation or image-enhancement postprocessing. It does not claim that arbitrary remote sensing datasets, architectures, finite-sample training procedures, or nonconvex optimizers automatically satisfy the assumptions. The VEDAI study is an assumption and risk diagnostic rather than a claim of end-to-end benchmark state-of-the-art.

II. Related Work

A. Remote Sensing Object Detection

Remote sensing object detection has long been recognized as a challenging problem because objects are embedded in large geospatial scenes, appear at multiple scales and orientations, and may be visually weak relative to structured background clutter [9]. Benchmarks such as DOTA and DIOR make these issues explicit through oriented aerial instances, large category sets, strong interclass similarity, and wide intraclass variation [10], [11]. Large overhead datasets such as xView emphasize geographic scale, fine spatial resolution, and many small annotated objects [12], while FAIR1M stresses fine-grained categories, oriented boxes, and high-resolution object variability [13]. General detectors such as Faster R-CNN and FPN provide anchor-based and multiscale templates that are widely adapted in remote sensing [14], [15]. However, sparse foregrounds make ordinary empirical risk misleading: an all-background predictor can have low raw error when positives are rare. This motivates class-balanced or task-calibrated objectives for the theoretical analysis. VEDAI provides coregistered color and near-infrared aerial imagery with oriented annotations for small vehicles under changes in orientation, illumination, shadow, and occlusion [16]. These properties make it a relevant real-data stress test for the local contrast and cross-modal assumptions studied here.

B. Multimodal AI and Remote Sensing Foundation Models

Recent multimodal remote sensing models emphasize transferable representations across sensors, scales, seasons, and regions. Cross-City Matters combines a hyperspectral–multispectral–SAR benchmark with domain-adaptive cross-city segmentation and dice-based imbalance treatment [17]; SeaMo learns season-aware optical–SAR features through masked pretraining and temporal–multimodal fusion [18]; and FlexiMo uses resolution-aware alignment and spectral-channel adaptation for heterogeneous inputs [19]. RSTPD instead asks which diffusion-time posterior is required for sparse detection and when auxiliary information lowers class-balanced Bayes risk. These encoders may instantiate Γ_i, but their representation objectives do not identify the equal-prior anchor posterior. RSTPD is therefore a complementary decision-risk layer rather than a competing backbone. Related methodological work also

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studies robustness and online adaptation. SAGG uses sample-adaptive gradient gating to handle heterogeneous multimodal corruption [20], while no-regret learning with hard real-time constraints and noise-adaptive high-probability regret bounds provide complementary tools for safe and statistically controlled online optimization [21], [22]. These works address optimization and training robustness rather than the task-posterior Bayes risk studied here, but they motivate future finite-sample, corrupted-modality, and streaming extensions of RSTPD.

C. Diffusion and Score-Based Modeling

Score matching estimates gradients of log densities without normalizing constants [23], and denoising score matching links this objective to corrupted observations [24]. Modern diffusion models use forward noising and reverse denoising to model complex image distributions [2], [3], [4]. Latent diffusion improves scalability for high-resolution synthesis [25]. These models learn distributions. Their primary score is tied to $p_t(x \mid z)$, which need not coincide with the posterior needed for a class-balanced decision.

D. Diffusion in Remote Sensing

Remote sensing diffusion work now covers controllable image generation, text-to-image data construction, synthetic data for detection, superresolution, fusion, and feature extraction. Liu et al. [1] survey this area in TGRS and identify task-specific interpretation and multimodal remote sensing diffusion as important future directions. CRS-Diff studies controllable remote sensing image generation [5]; MMM-RS provides a multimodal, multi-GSD text-to-image benchmark [6]; AeroGen uses diffusion-based data generation to improve remote sensing object detection [7]; and RS-Dseg uses a diffusion component for unsupervised feature extraction in segmentation [8]. These studies show that diffusion priors are useful, but they optimize image samples, generated training sets, restored images, or representation features. Here the central object is the class-balanced detection posterior. The question is not whether diffusion can improve images, but which posterior is needed for Bayes-optimal small sparse detection.

III. Problem Formulation

A. Valid Anchors and Class-Balanced Risk

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A remote sensing scene is represented by a clean main image or feature vector $X_0 \in \mathbb{R}^d$, auxiliary modalities $Z = (Z^{(1)}, \ldots, Z^{(K)})$, and a finite set of candidate anchors $\mathcal{I} = \{1, \ldots, N\}$. Anchor i has a binary target label $M_i \in \{0, 1\}$. Write $M = (M_1, \ldots, M_N)$ for the complete anchor-label vector, and let $d, d_0, \ldots, d_K$ denote positive integer feature dimensions. The sparse-mask assumption is

$$\sum_{i=1}^{N} M_i \le s \quad \mathbb{P}\text{-a.s} \tag{3}$$

where $s \in \{0, \ldots, N\}$ is the prescribed maximum number of target anchors in a scene. Only anchors with both classes present define a binary class-balanced problem

$$\mathcal{I}_+ = \{i : \mathbb{P}(M_i = 0) > 0,\ \mathbb{P}(M_i = 1) > 0\}, \quad N_+ = |\mathcal{I}_+|. \tag{4}$$

For $i \in \mathcal{I}_+$ and fixed diffusion time t, define the class-balanced measure $\mathbb{P}^{\circ}_{i,t}$ by

$$\mathbb{P}^{\circ}_{i,t}(A, M_i = m) = \frac{1}{2}\mathbb{P}(A \mid M_i = m), \quad m \in \{0, 1\}. \tag{5}$$

Under this measure, $\mathbb{P}^{\circ}_{i,t}(M_i = 0) = \mathbb{P}^{\circ}_{i,t}(M_i = 1) = 1/2$. For a detector h_i, the class-balanced risk is

$$R_{i,t}(h_i) = \mathbb{P}^{\circ}_{i,t}(h_i \ne M_i). \tag{6}$$

This risk penalizes missing positives even when the raw positive rate is extremely small.

B. Forward Diffusion and Local Statistics

Let T be a diffusion training time with probability law ρ on $\mathcal{T} \subset (0, 1]$, independent of (X_0, M, Z, G), and let

$$X_t = a_t X_0 + \sigma_t G, \qquad G \sim \mathcal{N}(0, I_d) \tag{7}$$

where $a_t \in (0, 1]$ and $\sigma_t > 0$, and I_q denotes the $q \times q$ identity matrix. For anchor i, let $P_i \in \mathbb{R}^{d_0 \times d}$ satisfy the whitened projection condition

$$P_i P_i^{\top} = I_{d_0}. \tag{8}$$

The main-image local feature is $V^{(0)}_{i,t} = P_i X_t$, and auxiliary local features are $V^{(k)}_i \in \mathbb{R}^{d_k}$ for $k = 1, \ldots, K$. The full local statistic is

$$V_{i,t} = \left(V^{(0)}_{i,t}, V^{(1)}_i, \ldots, V^{(K)}_i\right). \tag{9}$$

We assume $V_{i,t} = \Gamma_i(X_t, Z)$ for a measurable localization map Γ_i, so a detector using $V_{i,t}$ is a valid detector using (X_t, Z). When needed, we assume task sufficiency

$$\mathbb{P}^{\circ}_{i,t}(M_i = 1 \mid X_t, Z) = \mathbb{P}^{\circ}_{i,t}\left(M_i = 1 \mid V_{i,t}\right) \quad \mathbb{P}^{\circ}_{i,t}\text{-a.s.} \tag{10}$$

C. Local Gaussian Contrast Model

The clean main-image local contrast is modeled as follows:

$$V^{(0)}_{i,0} \mid (M_i = m) \sim \mathcal{N}\left(m\, u_i, \sigma^2_{b,i} I_{d_0}\right) \tag{11}$$

where $u_i \in \mathbb{R}^{d_0}$ and $\sigma_{b,i} > 0$. Auxiliary modality k satisfies

$$V^{(k)}_i \mid (M_i = m) \sim \mathcal{N}(m\delta_{ik}, \Sigma_{ik}), \quad \Sigma_{ik} > 0. \tag{12}$$

Unless stated otherwise, $V^{(0)}_{i,t}, V^{(1)}_i, \ldots, V^{(K)}_i$ are conditionally independent given M_i. This independence is used only to express information additively; a correlated Gaussian extension is given later. Table I summarizes the modeling assumptions, their mathematical roles, and how they can be interpreted or checked in remote sensing pipelines.

For later use, define the diffused main-image covariance scalar

$$\beta_{i,t} := a_t^2 \sigma^2_{b,i} + \sigma_t^2. \tag{13}$$

Thus, $\beta_{i,t} I_{d_0}$ is the within-class covariance of $V^{(0)}_{i,t}$.

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TABLE I
MAIN ASSUMPTIONS AND PRACTICAL DIAGNOSTICS

Assumption	Mathematical form	Role and practical diagnostic
Valid anchors	$i \in \mathcal{I}_+$, $\mathbb{P}(M_i = 0) > 0, \quad \mathbb{P}(M_i = 1) > 0$	Makes the class-balanced conditional measures well-defined. In practice, remove one-class anchors and report valid-anchor counts for each split.
Local sufficiency	$V_{i,t} = \Gamma_i(X_t, Z)$, $\mathbb{P}^\circ_{i,t}(M_i = 1 \mid X_t, Z) = \mathbb{P}^\circ_{i,t}(M_i = 1 \mid V_{i,t})$	Allows the local closed-form Bayes risk to represent complete-input risk when invoked. Check receptive fields, registration windows, and calibrated residual errors outside the local statistic.
Whitened projection	$P_i P_i^\top = I_{d_0}$	Makes diffusion noise isotropic in the local image feature. Approximate it by patch normalization, learned whitening, or covariance-normalized features.
Gaussian contrast	Equal class covariance with shifted class means	Gives an exact closed-form $\Phi(-\Delta/2)$ risk under the model and a diagnostic approximation otherwise. Check decision-score Q–Q plots, class-covariance mismatch, and held-out risk error.
Auxiliary information	$\Delta^2_{i,\mathrm{mod}} = \sum_k \delta_{ik}^\top \Sigma_{ik}^{-1} \delta_{ik} > 0$	Gives strict multimodal improvement over image postprocessing. Measure modality ablations and whitened target-background contrast for SAR, multispectral, elevation, temporal, or metadata features.
Cross-modal dependence	Block independence or joint covariance $C_i(t) \succ 0$	Independence is needed only for the additive contrast formula. Use the joint Mahalanobis expression for correlated Gaussian features and a flexible joint posterior for nonlinear dependence.

Definition 1 (Detector Classes): For fixed i, t, the image-postprocessing class is

$$\mathcal{H}_{\mathrm{img},i,t} = \left\{ h\left(V_{i,t}^{(0)}\right) : h : \mathbb{R}^{d_0} \to \{0,1\}\, h \text{ measurable} \right\}. \quad (14)$$

The multimodal task-posterior class is

$$\mathcal{H}_{\mathrm{all},i,t} = \left\{ h\left(V_{i,t}\right) : h : \mathbb{R}^{d_0+\cdots+d_K} \to \{0,1\}\, h \text{ measurable} \right\}. \quad (15)$$

Their optimal risks are

$$R^*_{i,\mathrm{img}}(t) = \inf_{h \in \mathcal{H}_{\mathrm{img},i,t}} R_{i,t}(h) \quad (16)$$

$$R^*_{i,\mathrm{all}}(t) = \inf_{h \in \mathcal{H}_{\mathrm{all},i,t}} R_{i,t}(h). \quad (17)$$

Define the image, auxiliary, and total detection information

$$\Delta^2_{i,\mathrm{img}}(t) = \frac{a_t^2 \|u_i\|_2^2}{a_t^2 \sigma_{b,i}^2 + \sigma_t^2} \quad (18)$$

$$\Delta^2_{i,\mathrm{mod}} = \sum_{k=1}^{K} \delta_{ik}^\top \Sigma_{ik}^{-1} \delta_{ik} \quad (19)$$

$$\Delta^2_{i,\mathrm{all}}(t) = \Delta^2_{i,\mathrm{img}}(t) + \Delta^2_{i,\mathrm{mod}}. \quad (20)$$

IV. DECISION-CALIBRATED TASK-POSTERIOR DIFFUSION

A. Class-Balanced Posterior Flow

Let

$$\eta_i^\circ(x,t,z) = \mathbb{P}^\circ_{i,t}(M_i = 1 \mid X_t = x, Z = z) \quad (21)$$

be a jointly measurable class-balanced posterior version. Define

$$\alpha^\circ_{i,1}(x,t,z) = \eta_i^\circ(x,t,z), \quad \alpha^\circ_{i,0}(x,t,z) = 1 - \eta_i^\circ(x,t,z). \quad (22)$$

If $p_t(x \mid z)$ denotes the conditional density of X_t given $Z = z$, the decision-calibrated task-posterior diffusion distribution is

$$p^\dagger_{i,t}(x, m \mid z) = p_t(x \mid z)\, \alpha^\circ_{i,m}(x,t,z), \quad m \in \{0,1\}. \quad (23)$$

The two posterior weights are nonnegative and sum to one, so this is a proper distribution. It is not the raw sparse-prior joint distribution; it is calibrated to the class-balanced decision problem in (5).

On regions where p_t and $\alpha^\circ_{i,m}$ are positive and differentiable

$$\begin{aligned} s^\dagger_{i,m}(x,t,z) &= \nabla_x \log p^\dagger_{i,t}(x, m \mid z) \\ &= \nabla_x \log p_t(x \mid z) + \nabla_x \log \alpha^\circ_{i,m}(x,t,z). \end{aligned} \quad (24)$$

The ordinary image score is $\nabla_x \log p_t(x \mid z)$. The task-posterior score is $\nabla_x \log \alpha^\circ_{i,m}(x,t,z)$. The risk advantage derived below comes from learning the posterior information in the task-posterior score, not merely from improving the image score.

B. Population Training Objective

The RSTPD population objective has two heads. A score head $s_\theta(x,t,z)$, with model parameters θ, is trained by denoising score matching

$$\mathcal{L}_{\mathrm{DSM}}(s) = \mathbb{E}\left[\left\| s(X_T, T, Z) + \frac{X_T - a_T X_0}{\sigma_T^2} \right\|_2^2 \right]. \quad (25)$$

A posterior head $\eta_\psi = (\eta_{\psi,i})_{i \in \mathcal{I}_+}$, with model parameters ψ, is trained by valid-anchor class-balanced cross-entropy. For $y \in \{0,1\}$ and $q \in [0,1]$, let

$$\ell(y,q) = \begin{cases} -\log q, & y = 1,\ q \in (0,1] \\ +\infty, & y = 1,\ q = 0 \\ -\log(1-q), & y = 0,\ q \in [0,1) \\ +\infty, & y = 0,\ q = 1. \end{cases} \quad (26)$$

The class-balanced posterior loss is

$$\mathcal{L}^\circ_{\mathrm{CE}}(\eta) = \sum_{i \in \mathcal{I}_+} \mathbb{E}_{\mathbb{P}^\circ_{i,\rho}}\left[\ell(M_i, \eta_i(X_T, T, Z)) \right] \quad (27)$$

where $\mathbb{P}^\circ_{i,\rho}$ is the mixture over $T \sim \rho$ induced by (5). The total objective is

$$\mathcal{L}_{\mathrm{RSTPD}}(s,\eta) = \mathcal{L}_{\mathrm{DSM}}(s) + \lambda \mathcal{L}^\circ_{\mathrm{CE}}(\eta), \quad \lambda > 0. \quad (28)$$

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Image-domain diffusion improves or scores the image before a separate detector is applied. RSTPD instead treats the posterior required by the detection decision as a learned object. During population training, a noisy sample $X_T = a_T X_0 + \sigma_T G$ is used for denoising score matching, while valid anchors contribute class-balanced cross-entropy terms for the posterior head. At inference, the learned posterior scores can be thresholded or decoded by the top-s rule. The consistency theorem below concerns the population objective, not finite-sample SGD.

V. Distribution-Free and Closed-Form Bayes-Risk Separation

Complete proofs of all theorems and lemmas in this article, including the auxiliary score, posterior, and measure-theoretic identities used in their derivations, are provided in the Supplementary Material.

Theorem 1 (Distribution-Free Multimodal Monotonicity): For fixed i, t, let $Q_{i,m}^{\text{img}}$ and $Q_{i,m}^{\text{all}}$ be the class-conditional laws of $V_{i,t}^{(0)}$ and $V_{i,t}$, respectively, under $M_i = m$. With the convention

$$\|P - Q\|_{\text{TV}} := \sup_A |P(A) - Q(A)| \tag{29}$$

the class-balanced Bayes risks satisfy

$$R_{i,\text{img}}^*(t) = \frac{1}{2}\left(1 - \left\|Q_{i,1}^{\text{img}} - Q_{i,0}^{\text{img}}\right\|_{\text{TV}}\right) \tag{30}$$

$$R_{i,\text{all}}^*(t) = \frac{1}{2}\left(1 - \left\|Q_{i,1}^{\text{all}} - Q_{i,0}^{\text{all}}\right\|_{\text{TV}}\right) \tag{31}$$

and therefore,

$$R_{i,\text{all}}^*(t) \le R_{i,\text{img}}^*(t). \tag{32}$$

The inequality is strict exactly when the joint multimodal class-conditional laws have strictly larger total-variation separation than their main-image marginals. No Gaussianity, equal covariance, density, or conditional-independence assumption is required.

Theorem 1 is the baseline guarantee for cluttered, non-Gaussian data and arbitrary linear or nonlinear cross-modal dependence. RSTPD does not impose a Gaussian posterior at training or inference: its posterior head is a joint function of (X_t, Z), and the population consistency result applies whenever that posterior is realizable. The Gaussian model below has the narrower role of converting the otherwise unknown total-variation increase into an interpretable closed form.

Lemma 1 (Gaussian Binary Risk): Let $W \mid (Y = m) \sim \mathcal{N}(m\mu, C)$, $m \in \{0, 1\}$, with $C \succ 0$, and let $\mathbb{P}(Y = 0) = \mathbb{P}(Y = 1) = 1/2$. Then the Bayes detector is

$$h^*(w) = \mathbf{1}\left\{\mu^\top C^{-1} w - \frac{1}{2}\mu^\top C^{-1}\mu \ge 0\right\} \tag{33}$$

and its risk is

$$R^* = \Phi\left(-\frac{1}{2}\sqrt{\mu^\top C^{-1}\mu}\right). \tag{34}$$

Theorem 2 (Closed-Form Risk Separation): Under the valid-anchor, forward-diffusion, whitened-projection, and local Gaussian assumptions, for any $i \in \mathcal{I}_+$ and fixed t

$$R_{i,\text{img}}^*(t) = \Phi\left(-\frac{1}{2}\Delta_{i,\text{img}}(t)\right) \tag{35}$$

$$R_{i,\text{all}}^*(t) = \Phi\left(-\frac{1}{2}\Delta_{i,\text{all}}(t)\right). \tag{36}$$

If the task-sufficiency condition (10) holds, then $R_{i,\text{all}}^*(t)$ is also the Bayes class-balanced risk among all detectors using (X_t, Z).

The same Gaussian calculation also covers correlated modalities by replacing the additive information with joint Mahalanobis information. If $V_{i,t} \mid (M_i = m) \sim \mathcal{N}(m\mu_i(t), C_i(t))$ with $C_i(t) \succ 0$, then

$$R_{i,\text{all}}^*(t) = \Phi\left(-\frac{1}{2}\sqrt{\mu_i(t)^\top C_i(t)^{-1}\mu_i(t)}\right). \tag{37}$$

This follows by applying Lemma 1 without the block-diagonal covariance calculation. Thus correlated modalities should be judged by their joint covariance-normalized contrast, not by a naive sum of marginal contrasts.

Theorem 3 (Strict Bayes Dominance): If $\Delta_{i,\text{mod}}^2 > 0$, then

$$R_{i,\text{all}}^*(t) < R_{i,\text{img}}^*(t). \tag{38}$$

With task sufficiency and population posterior consistency, RSTPD attains the multimodal risk and is strictly better than the optimal image-postprocessing detector at any certified time t.

Theorem 4 (Small-Target Nondegeneracy): Consider a target-size parameter $r \downarrow 0$ such that $\|u_i(r)\|_2 \to 0$, with $a_t, \sigma_t, \sigma_{b,i}$ fixed. Then

$$\lim_{r \downarrow 0} R_{i,\text{img}}^*(t; r) = \frac{1}{2}. \tag{39}$$

If there exists $b_i > 0$ such that $\Delta_{i,\text{mod}}^2(r) \ge b_i$ for all sufficiently small r, then

$$\limsup_{r \downarrow 0} R_{i,\text{all}}^*(t; r) \le \Phi\left(-\sqrt{b_i}/2\right) < \frac{1}{2}. \tag{40}$$

VI. Population Consistency

Assumption 1 (Regularity and Realizability): Assume that $p_t(x \mid z)$ exists, is positive and differentiable on the required region, and permits differentiation under the Gaussian convolution integral. Assume $\int_{\mathcal{T}} \sigma_t^{-2}\rho(dt) < \infty$, so the DSM target is square integrable. Assume also that the model class can realize both the true image score $\nabla_x \log p_t(x \mid z)$ and the class-balanced posterior $\eta_i^\circ(x, t, z)$ for all $i \in \mathcal{I}_+$.

Theorem 5 (RSTPD Population Consistency): Under Assumption 1, any global minimizer (s^*, η^*) of (28) in the realizable population function class satisfies

$$s^*(X_T, T, Z) = \nabla_x \log p_T(X_T \mid Z) \quad \text{a.s} \tag{41}$$

$$\eta_i^*(X_T, T, Z) = \mathbb{P}_{i,T}^\circ(M_i = 1 \mid X_T, Z) \quad \mathbb{P}_{i,\rho}^\circ\text{-a.s.} \tag{42}$$

For each $i \in \mathcal{I}_+$, there is a certified time set $\mathcal{T}_i^\star(\eta^*)$ with $\rho(\mathcal{T}_i^\star) = 1$ such that, for every $t \in \mathcal{T}_i^\star$, the plug-in detector

$$h_i^*(x, t, z) = \mathbf{1}\left\{\eta_i^*(x, t, z) \ge 1/2\right\} \tag{43}$$

attains the Bayes class-balanced risk using (X_t, Z). If (10) holds, this risk is the closed-form risk $R_{i,\text{all}}^*(t)$.

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A. Time Certification

Theorem 5 is an almost-everywhere statement over the training time measure ρ. Certifying a preselected time t_0 requires $\rho(\{t_0\}) > 0$ or an added fixed-time class-balanced CE term at t_0. A continuous-time distribution alone does not constrain a prescribed singleton time.

B. Posterior Values Versus Posterior Gradients

The CE term identifies posterior values on the data distribution. It does not, by itself, identify $\nabla_x \eta_i(x,t,z)$ pointwise. The learned task-posterior score in (24) is recovered only under an additional condition, such as pointwise C^1 equality of a selected posterior version on an open set, or an explicit Sobolev-type regularizer that constrains posterior gradients. Posterior values are enough for Bayes decisions; gradient-level score interpretation needs stronger regularity.

VII. Approximation and Robustness

Lemma 2 (Plug-In Excess Risk): Let $\eta(U) = \mathbb{P}(Y = 1 \mid U)$, $\hat{h}(U) = \mathbf{1}\{\hat{\eta}(U) \geq 1/2\}$, and $h^*(U) = \mathbf{1}\{\eta(U) \geq 1/2\}$. Then

$$R\left(\hat{h}\right) - R\left(h^*\right) \leq 2\mathbb{E}|\hat{\eta}\left(U\right) - \eta\left(U\right)|. \tag{44}$$

For the next result, define the complete-input Bayes risk

$$R^*_{i,\mathrm{Bayes}}\left(t\right) := \inf_{h:\,(X_t,Z)\mapsto\{0,1\}} R_{i,t}\left(h\right) \tag{45}$$

where the infimum is over measurable detectors.

Theorem 6 (Posterior Approximation Margin): For fixed i, t, let $U = (X_t, Z)$ and suppose

$$\mathbb{E}_{\mathbb{P}^\circ_{i,t}}|\widehat{\eta}_i\left(U\right) - \eta^\circ_i\left(U\right)| \leq \varepsilon_i\left(t\right). \tag{46}$$

Then

$$R_{i,t}\left(\widehat{h}_i\right) \leq R^*_{i,\mathrm{Bayes}}\left(t\right) + 2\varepsilon_i\left(t\right). \tag{47}$$

If task sufficiency holds and

$$2\varepsilon_i\left(t\right) < R^*_{i,\mathrm{img}}\left(t\right) - R^*_{i,\mathrm{all}}\left(t\right) \tag{48}$$

then the learned posterior detector is still strictly better than the optimal image-postprocessing detector.

Lemma 3 (Risk Stability in Total Variation): Let P_m, Q_m be class-conditional observation distributions, $m = 0, 1$. Let R_P and R_Q denote their corresponding equal-prior class-balanced risks. If $\|P_m - Q_m\|_{\mathrm{TV}} \leq \varepsilon_m$, then for any detector h

$$|R_P\left(h\right) - R_Q\left(h\right)| \leq \frac{1}{2}\left(\varepsilon_0 + \varepsilon_1\right). \tag{49}$$

Theorem 7 (Non-Gaussian Stability): Let $G^{\mathrm{img}}_m, G^{\mathrm{all}}_m$ denote the Gaussian class-conditional models used in Theorem 2, and let $Q^{\mathrm{img}}_m, Q^{\mathrm{all}}_m$ be the true class-conditional distributions. Suppose

$$\left\|Q^{\mathrm{img}}_m - G^{\mathrm{img}}_m\right\|_{\mathrm{TV}} \leq \zeta_m \tag{50}$$

$$\left\|Q^{\mathrm{all}}_m - G^{\mathrm{all}}_m\right\|_{\mathrm{TV}} \leq \varepsilon_m. \tag{51}$$

If the Gaussian Bayes gap

$$\gamma_G\left(t\right) = R^*_{G,\mathrm{img}}\left(t\right) - R^*_{G,\mathrm{all}}\left(t\right) \tag{52}$$

satisfies

$$\gamma_G\left(t\right) > \frac{1}{2}\left(\zeta_0 + \zeta_1 + \varepsilon_0 + \varepsilon_1\right) \tag{53}$$

then the true non-Gaussian risks still obey

$$R^*_{Q,\mathrm{all}}\left(t\right) < R^*_{Q,\mathrm{img}}\left(t\right). \tag{54}$$

VIII. Negative Result and Sparse Decoding

A. Pure Image Score Need Not Be Task-Informative

Theorem 8 (Image-Score Nonequivalence): There exists a model satisfying the valid-anchor, diffusion, local Gaussian, conditional independence, class-balance, and task-sufficiency assumptions such that X_t is independent of M_i. Hence every detector depending only on X_t, $P_i X_t$, or the pure image score $\nabla_x \log p_t(x)$ has class-balanced risk $1/2$. At the same time, an auxiliary modality with $\delta_i^\top \Sigma_i^{-1} \delta_i > 0$ gives a multimodal Bayes risk strictly below $1/2$.

B. Top-s Posterior Decoding

The risk formulas above are per valid anchor. At inference, many remote sensing systems require a sparse mask. Let U denote the available observation and $\eta_i(U) = \mathbb{P}(M_i = 1 \mid U)$. Under ordinary conditional Hamming risk with the pointwise constraint $\sum_i h_i(U) \leq s$, a Bayes-optimal decoder selects at most s anchors among those with $\eta_i(U) > 1/2$, choosing the largest posterior values. This follows because, conditional on $U = u$, the Hamming risk differs from a constant by:

$$\sum_i h_i\left(u\right)\left(1 - 2\eta_i\left(u\right)\right). \tag{55}$$

IX. Numerical Experiments and Real-Data Diagnostics

A. Analytic-Model Setup

We run three numerical checks under the analytic Gaussian anchor model used in Lemma 1. They are sanity checks for the risk formulas, the small-target limit, posterior calibration, and robustness margins, not remote sensing benchmark experiments. We then perform a separate real-data assumption diagnostic on VEDAI; the latter uses no samples generated from the analytic model.

Fig. 1(a) plots the strict risk gap $R_{\mathrm{img}} - R_{\mathrm{RSTPD}}$, which becomes positive once $\Delta^2_{\mathrm{mod}} > 0$. Fig. 1(b) shows the small-target limit: image-only error approaches $1/2$, while RSTPD remains near 0.25. Fig. 1(c) varies posterior logit noise and auxiliary contamination; the contour marks where the plug-in detector loses its positive margin. Fig. 1(d) gives clean and mismatched reliability diagrams.

B. Risk-Gap and Small-Target Checks

The strict-dominance check isolates the Bayes gap in Theorem 3. When $\Delta^2_{\mathrm{mod}} = 0$, image-only and multimodal risks coincide; once auxiliary Mahalanobis information is positive, the heatmap shows the predicted positive gap. The first row of Table II gives the same effect by Monte Carlo: image-only error is 0.345, RSTPD error is 0.209, and the gap is 0.136.

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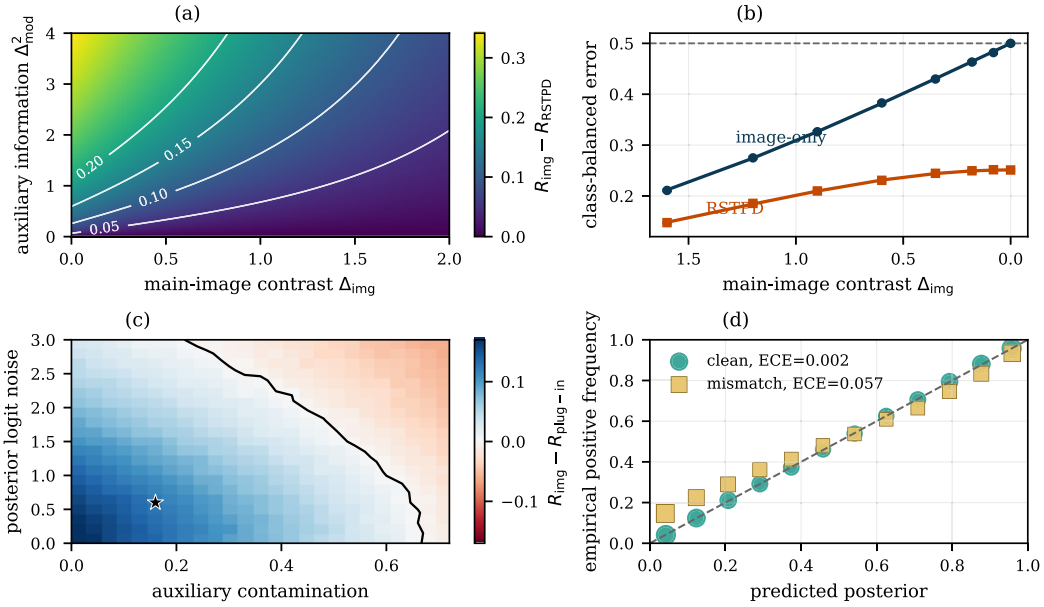


Fig. 1. Core numerical experiments. (a) Strict risk-gap heatmap under the Gaussian model. (b) Small-target limit with Monte Carlo markers and 95% intervals. (c) Robust posterior margin under logit noise and auxiliary contamination; the star marks the setting in Table II. (d) Reliability diagrams for clean and mismatched posteriors.

TABLE II

CORE NUMERICAL EXPERIMENTS UNDER THE ANALYTIC GAUSSIAN ANCHOR MODEL. ALL ENTRIES USE A FIXED RANDOM SEED; MC DENOTES MONTE CARLO CLASS-BALANCED ERROR

Claim tested	Diagnostic setting	Image MC	RSTPD MC	Gap / ECE	Verdict
Strict Bayes dominance	$\Delta_{\text{img}} = 0.80$ $\Delta^2_{\text{mod}} = 2.00$	0.345	0.209	0.136	Auxiliary Mahalanobis information gives a strict risk gap
Small-target non-degeneracy	$\Delta_{\text{img}} = 0$ $\Delta^2_{\text{mod}} = 1.80$	0.500	0.251	0.249	Image-only degenerates to random guessing while RSTPD remains below $1/2$
Posterior robustness	logit noise $= 0.60$ contamination $= 0.16$	0.383	0.265	0.118	Positive margin remains under moderate approximation and model mismatch
Posterior calibration	clean posterior, 12 reliability bins	0.383	0.211	ECE 0.002	Population posterior is empirically calibrated in the model

Because the samples are class-balanced, the gap comes from the auxiliary information term rather than class imbalance.

The small-target check probes Theorem 4. Main-image contrast is driven to zero while the auxiliary term is fixed. The image-only curve and table row reach 0.500, the balanced risk of random guessing, whereas RSTPD remains at 0.251. Thus, a vanishing main-image target does not force posterior degeneration when a registered modality retains target information.

C. Robustness and Calibration

The robustness panel relaxes the population calculation by adding posterior logit noise and auxiliary contamination. The positive region in Fig. 1(c) and the 0.118 gap in Table II show that the posterior advantage is not tied to one exact clean posterior.

The calibration row separates risk improvement from probability calibration. The clean model has expected calibration error (ECE) 0.002, so posterior values match empirical conditional frequencies in the analytic setting. Mismatch visibly degrades reliability, but Fig. 1(c) shows that a positive risk margin can remain. This supports the intended reading: posterior learning gives the decision advantage, while calibration and robustness diagnostics show when the population statement remains informative.

D. Vedai Real-Data Protocol

VEDAI provides co-registered RGB–NIR aerial images with oriented small-vehicle annotations [16]. Using the official 512 × 512 release and ten scene-level folds, we treat RGB as the main image and NIR as auxiliary. Each retained target is paired with a nonoverlapping, size- and orientation-matched background anchor from the same scene; excluding 482 boundary-crossing targets leaves 3212 anchors per class from 1118 scenes. Preprocessing is fit on nine folds and fixed for the held-out fold. Orientation-aligned 32 × 32 windows at 1.8 times the annotated extent yield pooled mean,

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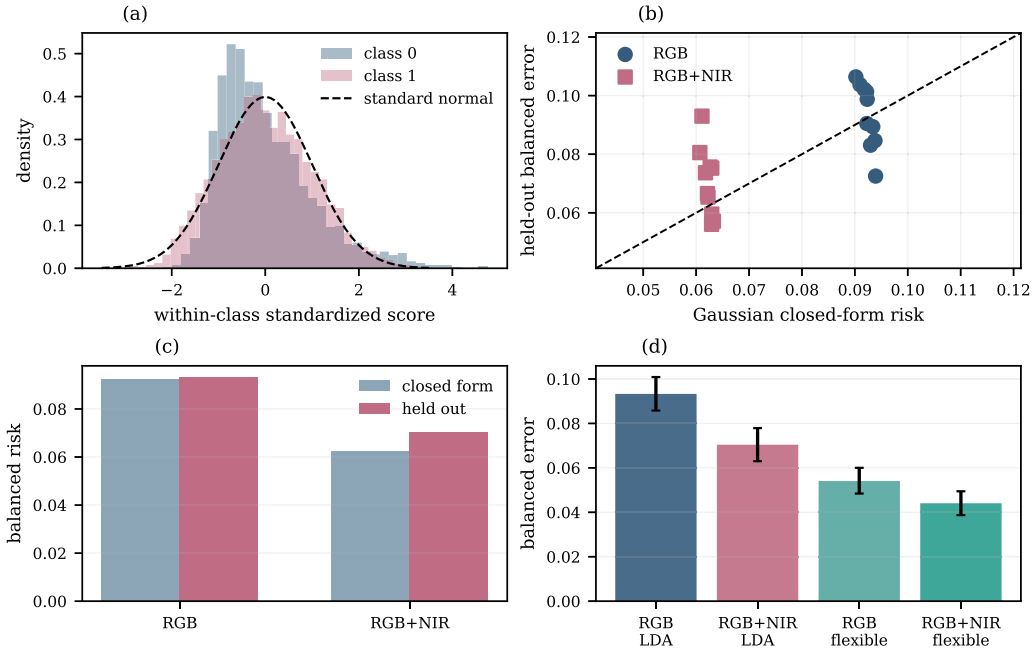


Fig. 2. Real-data validation of decision-level risk predictions on VEDAI. (a) Held-out class-conditional linear-discriminant scores compared with a standard normal density after within-class standardization. (b) Training-fold closed-form risks versus balanced errors on the held-out scene fold. (c) Aggregate agreement between closed-form and held-out risks. (d) Scene-bootstrap balanced errors for joint-covariance linear and flexible non-Gaussian posteriors; bars show means and error bars show 95% confidence intervals.

standard-deviation, and gradient descriptors, whitened to ten RGB and five NIR components. These fixed features test the local abstraction rather than an end-to-end detector. We assess held-out discriminant-score Gaussianity by within-class Q–Q R^2, compare training-fold closed-form risk with scene-disjoint balanced error for RGB and full-covariance RGB–NIR, include a histogram-gradient-boosting non-Gaussian posterior, and obtain confidence intervals from 2000 scene-level bootstrap resamples. The results are summarized in Table III.

E. Vedai Findings

The real-data results support the Gaussian model as a decision-relevant risk approximation. Fig. 2 visualizes these diagnostics. The joint discriminant score has Gaussian Q–Q $R^2 = 0.929$. More importantly, the full joint-covariance closed-form risk is 0.062, close to the 0.070 balanced error on unseen scenes; RGB alone gives 0.092 and 0.093, respectively. The observed RGB-to-joint error reduction is 0.023, with scene-bootstrap 95% CI $[0.017, 0.029]$. Thus, the formula correctly predicts both the ordering and the practical scale of the multimodal advantage.

The positive result persists without a Gaussian classifier. The flexible posterior gives errors 0.054 for RGB and 0.044 for RGB–NIR, with a reduction CI of $[0.006, 0.014]$. Together, these results show that the full-covariance Gaussian abstraction is useful at the risk level and that auxiliary information remains beneficial under a nonlinear non-Gaussian posterior. The score-projection and risk agreement validate an approximation rather than asserting exact multivariate Gaussianity; the distribution-free Theorem 1 covers the latter distinction.

F. Componentwise Ablation

Table IV consolidates one componentwise ablation from the existing evaluations. The Bayes-risk row isolates the sparse-label correction: at 1% target prevalence, an all-background rule appears to have only 0.010 raw error but has 0.500 class-balanced risk. This is not a performance gain by itself; it shows why the proposed objective exposes a failure hidden by raw accuracy. The task-posterior row isolates the decision mechanism in the analytic small-target limit, where the pure image/image-score decision is random while the multimodal posterior remains informative. The two VEDAI rows isolate linear full-covariance and flexible nonlinear fusion. Both confidence intervals exclude zero, so the measured fusion gains are not explained by fold-to-fold scene sampling alone.

G. Reproducibility and Quality Control

Table II and Fig. 1 use a fixed seed, with 180 000 balanced samples per Monte Carlo setting, 70 000 per robustness cell, and 220 000 for calibration. Checks enforce finite nondegenerate metrics, valid risk ranges, class balance, closed-form agreement, small-target nondegeneracy, the same balanced law for image-only baselines, and both positive- and negative-margin robustness regimes. The VEDAI diagnostic uses seed 20 260 902, official images, annotations, and folds, and `run_vedai_assumption_diagnostics.py`; it verifies paired RGB/NIR availability, scene-disjoint testing, balanced matched anchors, finite features, and complete out-of-fold predictions.

H. Computational Complexity and Efficiency

RSTPD fixes the posterior target rather than the backbone, so hardware-specific totals depend on the chosen implementation. For a backbone with P_b parameters, F_b MACs, and A_b activation words, let $d_v = \sum_{k=0}^{K} d_k$ and let the shared posterior

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TABLE III

REAL-DATA VALIDATION OF DECISION-LEVEL RISK PREDICTIONS ON THE VEDAI RGB–NIR SMALL-VEHICLE BENCHMARK. VALUES ARE AGGREGATED OVER THE OFFICIAL TEN SCENE-LEVEL FOLDS; CI DENOTES A 95% SCENE-BOOTSTRAP CONFIDENCE INTERVAL

Diagnostic	RGB only	RGB+NIR	Interpretation
Decision-score Gaussian Q–Q R^2	0.911	0.929	The high joint value supports Gaussian approximation in the decision-relevant score projection.
Closed-form / held-out balanced error	0.092 / 0.093	0.062 / 0.070	The full joint-covariance formula closely predicts risk on unseen scenes.
Joint-covariance risk reduction	–	0.023 [0.017, 0.029]	The positive CI verifies a scene-level multimodal advantage under the Gaussian risk approximation.
Flexible posterior balanced error	0.054	0.044	A non-Gaussian discriminative posterior tests whether auxiliary information remains useful without an equal-covariance restriction.
Flexible-posterior risk reduction	–	0.010 [0.006, 0.014]	The positive CI confirms that the multimodal gain persists beyond the Gaussian classifier.

TABLE IV

ONE CONSOLIDATED COMPONENTWISE ABLATION USING THE EXISTING ANALYTIC AND VEDAI EVALUATIONS. LOWER RISK OR ERROR IS BETTER. THE 1% PREVALENCE ROW IS AN ILLUSTRATIVE SPARSE-PRIOR CALCULATION, NOT AN ADDITIONAL DATASET EXPERIMENT

Component isolated	Without component	With component	Quantitative contribution
Class-balanced Bayes formulation at 1% target prevalence	All-background raw error: 0.010	All-background balanced risk: 0.500	Exposes a 0.490 hidden foreground penalty that raw accuracy suppresses.
Task-posterior mechanism, analytic small-target limit	Pure image/image-score risk: 0.500	Multimodal posterior risk: 0.251	Avoids posterior degeneration; absolute risk reduction 0.249.
Joint-covariance multimodal fusion, VEDAI	RGB error: 0.093	RGB+NIR error: 0.070	Reduction 0.023, with 95% CI [0.017, 0.029].
Flexible nonlinear multimodal fusion, VEDAI	RGB error: 0.054	RGB+NIR error: 0.044	Reduction 0.010, with 95% CI [0.006, 0.014].

MLP have widths $h_0 = d_v, \ldots, h_{L+1} = 1$. Its incremental costs are

$$P_\eta = \sum_{\ell=0}^{L} (h_\ell + 1)\, h_{\ell+1}, \qquad F_\eta = N_+ \sum_{\ell=0}^{L} h_\ell h_{\ell+1}. \tag{56}$$

Total parameters, MACs, and activation memory are $P_b + P_\eta$, $F_b + F_\eta$, and $A_b + O(N_+ \max_\ell h_\ell)$, respectively; anchorwise cost is linear and can be batched or tiled. For the VEDAI dimension $d_v = 15$, a shared 15–64–1 head has 1089 parameters, 1024 MACs per anchor, 4.25 KiB of FP32 weights, and 256 KiB of hidden activations for 1024 anchors. Certified-time inference requires one backbone/head pass rather than reverse diffusion sampling. The offline full-covariance fit costs $O(nd_v^2 + d_v^3)$ time and $O(d_v^2)$ storage, followed by $O(d_v)$ work per anchor; exact latency and peak memory remain encoder- and hardware-dependent.

I. Limitations, Scalability, and Future Work

The analytic experiments verify the formulas, while the VEDAI study tests the assumptions and risk approximation on real data. However, VEDAI uses fixed local descriptors and matched-anchor classification rather than an end-to-end detector with AP evaluation. Moreover, Theorem 1 guarantees only weak Bayes dominance beyond the Gaussian and independence assumptions; a strict quantitative gain still requires distribution-specific information or empirical validation.

The guarantees are local, population-level, and conditional on valid anchors and task sufficiency; they do not establish finite-sample complexity, nonconvex convergence, open-set performance, or robustness to arbitrary label noise, registration error, missing modalities, and sensor bias. Future work should combine RSTPD with scalable encoders such as FlexiMo and SeaMo, derive finite-sample and domain-shift bounds, address missing, asynchronous, or misregistered modalities, and evaluate end-to-end detectors across datasets, sensors, and regions using AP/mAP, calibration, modality-dropout and diffusion-time ablations, throughput, memory, and energy. Shared heads, sparse routing, low-rank fusion, distillation, and early exit are promising routes to large-scene deployment.

X. CONCLUSION

We developed a Bayes-risk theory for decision-calibrated task-posterior diffusion in remote sensing that separates image-score learning from the class-balanced posterior required by sparse small-object detection. Under the local Gaussian model and its TV-stable extension, auxiliary Mahalanobis information yields strict improvement, prevents degeneration when main-image contrast vanishes, and supports population consistency, posterior approximation margins, a pure image-score negative result, and sparse top-s decoding. Beyond the Gaussian calculation, total-variation contraction establishes distribution-free weak multimodal dominance, while the VEDAI diagnostic provides positive real-data support: full joint-covariance risk closely tracks held-out error, and both linear and flexible joint posteriors improve over their RGB counterparts. RSTPD therefore complements representation-centric multimodal foundation models by supplying a decision target and risk guarantee for a chosen encoder; the

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consolidated ablation isolates sparse-risk calibration, task-posterior information, and fusion gains, and the complexity analysis shows linear posterior-head scaling with the number of valid anchors. The claims remain local and population-level rather than finite-sample or state-of-the-art guarantees; future work should evaluate end-to-end architectures across multiple datasets, regions, sensor combinations, missing or misregistered modalities, and diffusion times while reporting detection accuracy, calibration, throughput, and memory.

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